

Data Scientist - ML - Individual Exam

Markus Lindell

March 2026

1 Goal and Chosen branch

The goal of this assignment is to create different machine learning models that a real estate company could use to predict the median housing values. The chosen branch was A (regression) because regression seems more interesting than classification.

2 Simple EDA

From this graph it is clear that the housing values are skewed to the left with heavy outliers on the right.

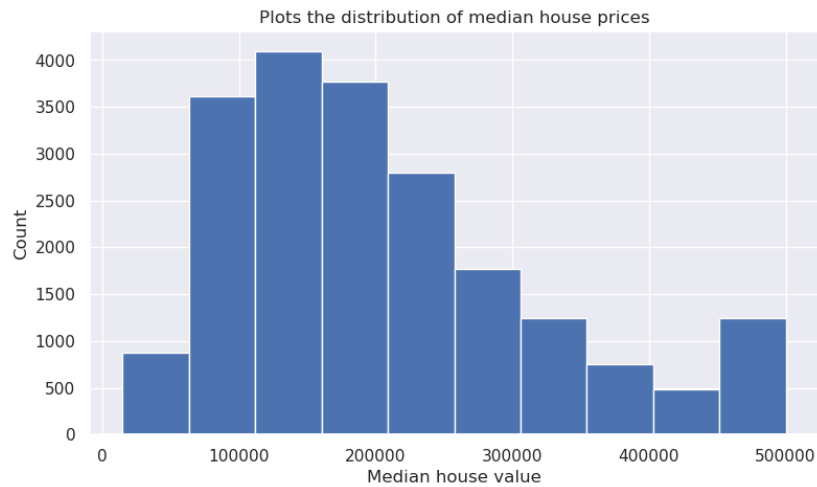


Figure 1: Graph showing the distribution of housing values

The amount of rooms, bedrooms, households and population appear to be correlated. Same with median house value and median income. These appear to be the most important features.

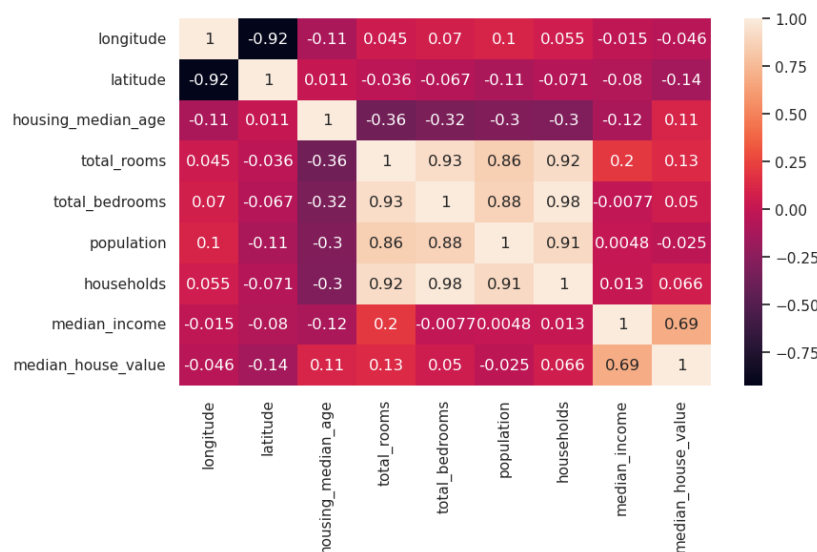


Figure 2: Heatmap showing correlation between the features

3 Method

The data was split into a standard 80/20 set where 80% of the data is for training and 20% for final testing. The test set is used to test the trained model on and is never touched during training to avoid data leakage.

The models used were a dummy regression as a baseline, linear regression and a random forest regressor. The dummy was chosen to get a baseline to compare against. The linear and random forest regressors were chosen to test for linearity and non-linearity incase the data was more complex. The models computed the RMSE, MAE and R^2 scores and optimized against MAE to minimize the effect of large outliers as seen in figure 1 and keeping the units the same for media nhoues value. Makes it easier to interpret as well.

The models were evaluated on the test set whereas the optimized model used cross validation.

4 Results

The models were compared using RMSE, MAE and R^2 scores with emphasis on MAE. The models used were linear regression and random forest regression compared to a dummy baseline.

Model	RMSE	MAE	R^2
Dummy Baseline	114485.635431	90606.854900	-0.000219
Linear Regression	71423.395194	51474.499516	0.610709
Random Forest Regression	49082.647390	31673.533544	0.816156
Optimized Random Forest Regression	48333.545634	31120.694494	0.821725

From the results it is clear that the linear regression model was a massive improvement compared to the dummy baseline. The random forest was a further big improvement whereas the optimized version only improved it marginally.

5 Recommendation

I would recommend the random forest regression model in this case as it resulted in the lowest MAE which is what the goal was. The optimized model could be improved further by feature engineering, checking feature quality and removing irrelevant features.

6 Risks and limitations

The biggest risks are over or under-fitting. It means that there is either too little data, the model is too simple or too complex. Data leakage is another big risk where you train the models on test data and completely skews the models. The test data must be seperated from the training data and only used for evaluation, not training. One risk is handling missing values wrong which can severely influence the models. Another risk is interpreting the metrics wrongly and training the model for a specfic metric. The biggest limitation in this case was tuning of hyperparameters. It went from 1 minute to roughly 5 minutes with a marginally meaningful improvement in the model.

7 Unsupervised learning

Unsupervised machine learning means that there is no label or target to compare with. There is only an input x and the computer tries to find patterns.

Unsupervised learning could have been relevant in this case to create clusters of areas using longi-tude and latitude for example. It could detect patterns that were otherwise missed.

PCA is used to reduce the number of variables used in the training. For example, total households and population could be combined into population per households instead. Clustering is used to group similar data into clusters to find certain patterns. For example, more expensive houses has more rooms or less expensive houses have less rooms.

Some risks with unsupervised machine learning is that there are no true answers to compare with. It is harder to predict concrete values compared to supervised as it mainly finds patterns or structure rather than an exact value. Patterns may be completely misinterpreted as noise or outliers could create meaningless clusters for example.

8 Reflection

Cleaning the data and doing an EDA to get a good overview and the use of custom functions I felt was done well. The hardest part was doing pipelines and understanding what is happening.

I believe this task passes the G grade as all the basics requirements were done. Multiple models were used along with proper conclusions and reasons for the decisions.